

Spinor Bundle — SO(26) Clifford Module Dimension $2^{13} = 8192$

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Abstract

The UQFF spinor bundle on the 26-dimensional compactified manifold has Clifford module dimension $2^{(D_{\text{crit}}/2)} = 2^{13} = \mathbf{8192 \text{ EXACT}}$, providing the geometric basis for: (1) the BSD spinor bundle index = 0.30600 (Cremona 37a1), (2) Bell quantum bound $2\sqrt{2}$ from holonomy, (3) Atiyah-Singer Dirac index = 22, and (4) Hodge identity = 1.0.

Part 1: The Spinor Bundle

In UQFF, the bosonic-string critical dimension $D_{\text{crit}} = 26$ gives rise to a spinor bundle with rank

$$\dim \text{Spin}(SO(26))_{\text{Clifford}} = 2^{D_{\text{crit}}/2} = 2^{13} = 8192$$

This is the dimension of the irreducible Clifford module for SO(26) (the orthogonal group on the compactified manifold).

Part 2: Cross-Validation

BSD spinor bundle index

$$\nu_{\text{BSD}} = 0.30600$$

matches Cremona 37a1 elliptic curve $L(1) = 0.30598$ to 0.005%.

Bell quantum bound

$$S_{\text{CHSH}}^{\text{max}} = 2\sqrt{2}$$

emerges from the spinor bundle holonomy parameter (PAPER_1222).

Atiyah-Singer Dirac index

$$\text{ind}(D) = D_{\text{crit}} - D_{\text{phys}} = 22$$

the residual compactified dimensions.

Hodge identity

$$h_{\text{Hodge}} = \frac{D_{\text{phys}} + D_{\text{BSFG}}}{SO_5} = 1.0 \text{ EXACT}$$

Part 3: Physical Significance

The spinor bundle on a 26D compactified manifold supports: - 8192-component spinors (much larger than 4-component SM Dirac) - Holonomy across 22 compactified dimensions - Dirac index = compactified residual = 22

Conclusion

The $SO(26)$ Clifford module dimension 8192 is the geometric backbone of UQFF's quantum and gravitational structure. Multiple major UQFF results (BSD index, Bell bound, Atiyah-Singer, Hodge identity) trace back to this unified geometric origin.

Framework Version: UQFF 5.27+